PHY 1002	Game Physics	L,T 2,1

Objective

Game Physics is a course for the students to

- learn the required basics of physics needed for game programming
- Train the students to add realism to the games scientifically during the game development process.

Expected outcome

- To develop realistic games using laws of physics
- To meet the need of gamers that demand quite more realistic approach.
- To improve game development skills by independent conceptual thinking.
- To increase the desire for the game programmers in creating fascinating games.

Unit	Topics	Lec Hrs	SLO
1	**Basic concepts** Game physics – game engine (introduction) - Physics realism – importance in games, physics concepts and game performance, Fundamentals - Coordinate Systems and Frames of Reference, Scalar and vectors, computing magnitude of vector, vector cross product, matrices - multiplication and rotation, derivatives,, Exercise and problem solving.	5	a,e
2	**Basic Newtonian mechanics & Kinematics** Newtons three laws of motions – inertia – force – mass – acceleration-equal & opposite forces , Force vector, Types of forces – gravitational – friction-centripetal – force balance and diagrams, Work, Energy – kinetic – potential – conservation – power, Translational motion – equation of motion, Rotational motion - torque – angular acceleration, 2D particle kinematics, 3D particle kinematics, Rigid body kinematics – centre of mass – coordinate axis – rolling motion –- bowling ball kinematics, Exercise and problem solving.	10	a,b
3	**Projectiles** Projectile properties, Simple trajectories & gravity, Drag –drag coefficient – drag effects and equation of motion, Wind effects to projectile, Magnus effect – spin effect to projectile, specific projectiles types in games – cannon balls – bullets – arrows, Exercise and problem solving.	6	b,h
4	**Collisions** Impulse and momentum principle- linear and angular, elastic & inelastic collisions Impact, coefficient of restitution, collision direction and detection, collision with movable & immovable objects, collision with friction, 2D and 3D collisions, Application to games, Exercise and problem solving.	9	e,h
5	**External Physics** External physics of gaming vehicles (motorcycles, cars, ships, airplanes, rocket, missiles) - basic force diagram – acceleration - braking - travel around curves - drag - lift –gravity – orbit – thrust - buoyancy & balance - take off & landing and other game building related concepts, Physics of sound – nature - behaviour – properties – sound synthesis for games Exercise and problem solving.	11	j,k
6	**Contemporary Topics & Guest Lectures**	4	e,k
Total Lecture hours		45	
Text Books			
1	Physics for Game Programmers, Grant Palmer, Apress publishers, (2005)		

2	Fundamental Physics, Halliday – Resnick, 8th Edtion, Weiley (2009)
3	Game Physics, David H. Eberly, Morgan Kaufmann publications, Second Edition (2010)
4	Classical Mechanics, Herbert Goldstein, 3rd Edition, Addison- Wesley, (2002).
Reference Books	
1	Game Physics Engine Development, Ian Millington, Morgan Kaufmann Publishers (2007), 2nd Edition, CRC press, 2010
2	Mathematics and Physics for Programmers, John Patrick Flynt & Danny Kodicek, Second Edition, Course Technology, a part of Cengage Learning. (2012).
3	Mechanics, Keith R. Symon, 3rd Edition, Addison- Wesley, (1971)
4	Advanced Physics, M. Nelkon, P. Parker, 7th Edition, Heinemann Educational Books (1970).

Recommendation by the board of studies	26-06-18
Approved by ACM	18-07-18